UNIT 3 INTEGRATION

5.2 The Definite Integral

- **Riemann Sums**

  Consider a function defined on a closed interval $[a, b]$. Divide $[a, b]$ into n equal subintervals by means of points $a = x_0 < x_1 < x_2 < \ldots < x_{n-1} < x_n = b$, where $x_i = a + i\Delta x$ and $\Delta x = \dfrac{b-a}{n}$. On each subinterval $[x_{i-1}, x_i]$, pick an arbitrary point x_i^*, which we call a sample point of the interval.

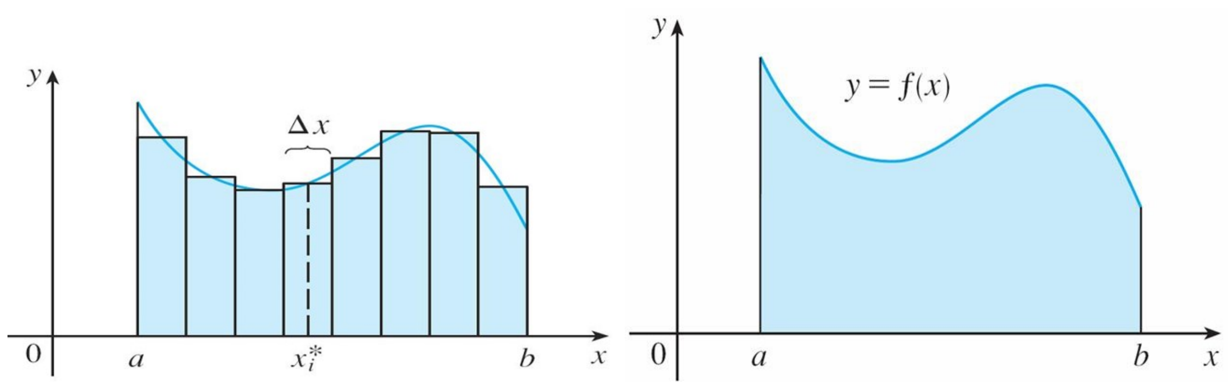
Let f be a function that is defined on the closed interval $[a, b]$. If

$$\lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i^*\right) \Delta x$$

exists, we say that f is **integrable** on $[a, b]$, and denote this limit by $\int_a^b f(x)dx$, called the **definite integral** of f from a to b. Namely,

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(x_i^*\right) \Delta x$$

a and b are called the **lower limit** and **upper limit** of the integral, $f(x)$ is called the **integrand**.



- **Example** Evaluate $\int_0^3 \left(2x^2 - 8x\right)\, dx$.

Special Sum Formulas

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

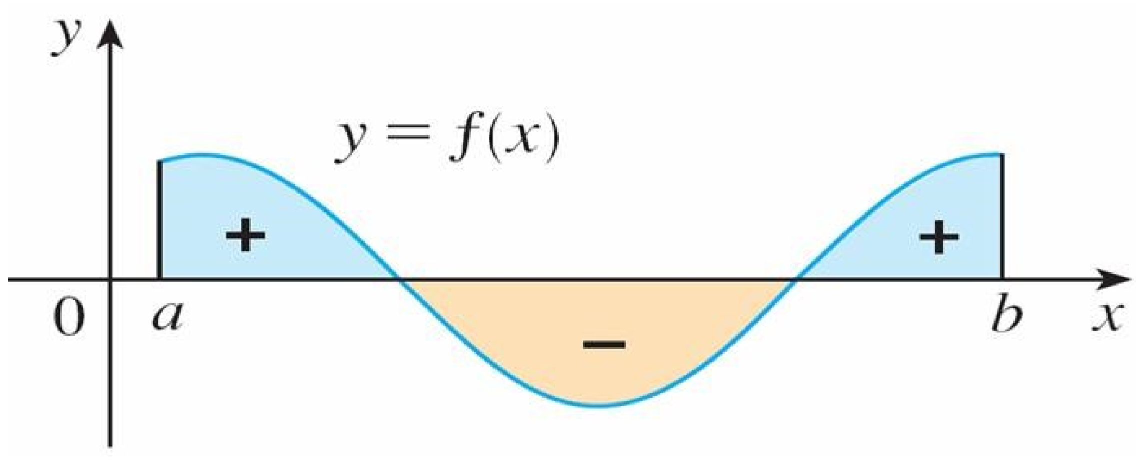
- **Theorem**

If f is bounded on $[a, b]$ and if it is continuous there except at finite number of points, then f is integrable on $[a, b]$.

If f is continuous on $[a, b]$, then it is integrable on $[a, b]$.

- **Geometric Interpretation**

$$\int_a^b f(x)dx = A_{\text{up}} - A_{\text{down}}$$



- **Example** Evaluate $\int_0^3 \sqrt{9-x^2}dx$ using plane geometry.

- **Properties of the Definite Integral**

> - $\int_a^a f(x)dx = 0.$
> - $\int_a^c f(x)dx + \int_c^b f(x)dx = \int_a^b f(x)dx.$
> - $\int_a^b f(x)dx = -\int_b^a f(x)dx.$

- **Example** Given that $\int_0^{10} f(x)dx = 17$, $\int_0^8 f(x)dx = 12$, find $\int_8^{10} f(x)dx$.

- **Linearity of the Definite Integral**

If f and g are integrable on $[a, b]$ and k is a constant, then kf and $f \pm g$ are integrable and

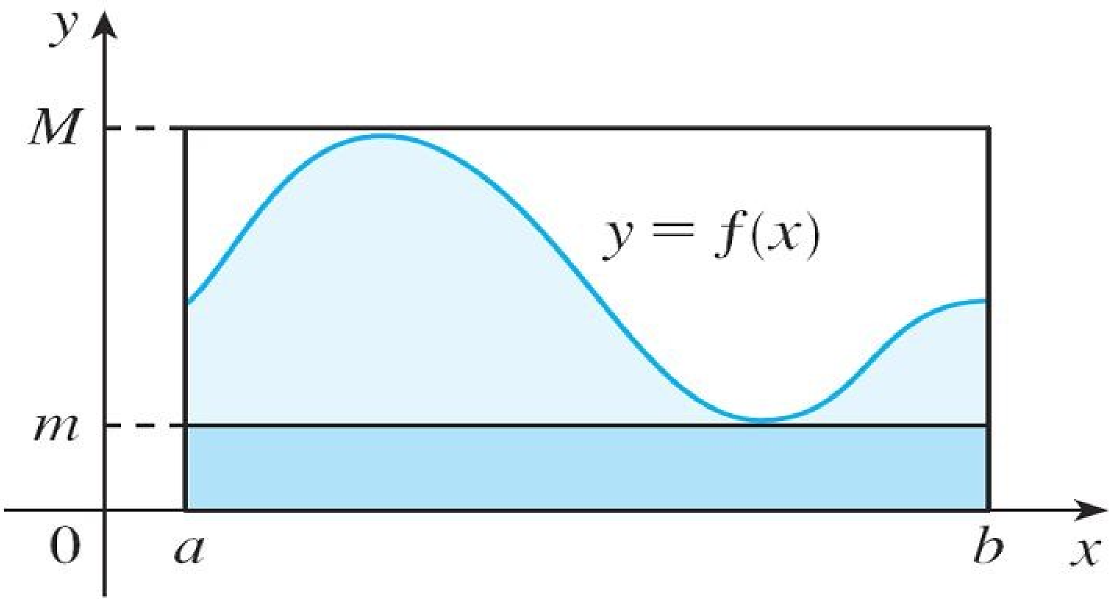
- $\displaystyle\int_a^b kf(x)dx = k\int_a^b f(x)dx$
- $\displaystyle\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$

- **Comparison Property**

If f and g are integrable on $[a,b]$ and if $f(x) \le g(x)$ for all $x \in [a,b]$, then
$$\int_a^b f(x)dx \le \int_a^b g(x)dx$$

- **Boundedness Property**

If f is integrable on $[a,b]$ and $m \le f(x) \le M$ for all x in $[a,b]$, then
$$m(b-a) \le \int_a^b f(x)dx \le M(b-a)$$



Exercises 5.2 17, 19, 29, 35, 37, 39, 41, 42, 43, 47, 49, 51, 53.

5.3 The Fundamental Theorem of Calculus

- **First Fundamental Theorem of Calculus**

Let f be continuous on the closed interval $[a, b]$ and let x be a variable point in (a, b). Then

$$\frac{d}{dx}\int_a^x f(t)dt = f(x)$$

- **Example** Find $\dfrac{d}{dx}\left[\displaystyle\int_1^x t^3 dt\right]$.

- **Example** Find $\dfrac{d}{dx}\left[\displaystyle\int_x^4 \tan^2 u \cos u du\right]$, $\dfrac{\pi}{2} < x < \dfrac{3\pi}{2}$.

- **Example** Find $D_x\left[\displaystyle\int_x^{x^2} \sqrt{3t^2-1} dt\right]$.

- **Second Fundamental Theorem of Calculus**

> Let f be continuous on $[a, b]$, and let F be any antiderivative of f on $[a, b]$.
>
> Then
> $$\int_a^b f(x)dx = F(b) - F(a)$$

- **Example** Evaluate $\int_{-1}^{2} \left(4x - 6x^2\right) dx$.

- **Example** Evaluate $\int_1^8 \left(x^{\frac{1}{3}} + x^{\frac{4}{3}}\right) dx$.

- **Example** Find $\int_0^x 3\sin t dt$.

- **Example** Evaluate $\int_1^3 e^x dx$.

- **Example** Evaluate $\int_3^6 \frac{1}{x} dx$.

- **Example** Find the area under the curve $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{2}$.

Exercises 5.3 7, 11, 13, 17, 19, 21, 23, 25, 29, 31, 45, 51.

5.4 Indefinite Integrals and Definite Integrals

- **Standard Integral Formulas**

1. $\int kdu = ku + C$

2. $$\int u^r du = \frac{u^{r+1}}{r+1} + C, \quad r \neq -1$$

3. $$\int \frac{1}{u} du = \ln |u| + C$$

4. $$\int e^u du = e^u + C$$

5. $$\int a^u du = \frac{a^u}{\ln a} + C, \quad a > 0,\ a \neq -1$$

6. $$\int \sin u du = -\cos u + C$$

7. $$\int \cos u du = \sin u + C$$

8. $$\int \sec^2 u du = \tan u + C$$

9. $$\int \csc^2 u du = -\cot u + C$$

10. $$\int \sec u \tan u du = \sec u + C$$

11. $$\int \csc u \cot u du = -\csc u + C$$

12. $$\int \frac{1}{\sqrt{1-u^2}} du = \sin^{-1} u + C$$

13. $$\int \frac{1}{1+u^2} du = \tan^{-1} u + C$$

14. $\displaystyle\int \sinh u du = \cosh u + C$

15. $\displaystyle\int \cosh u du = \sinh u + C$

- **Example** Evaluate $\displaystyle\int \left(10x^4 - 2\sec^2 x\right) dx$.

- **Example** Evaluate $\displaystyle\int \frac{\cos\theta}{\sin^2\theta} d\theta$.

- **Example** Evaluate $\displaystyle\int_0^2 \left(2x^3 - 6x + \frac{3}{x^2+1}\right) dx$.

- **Example** Evaluate $\displaystyle\int_1^9 \frac{2t^2 + t^2\sqrt{t} - 1}{t^2} dt$.

Exercises 5.4 5, 7, 9, 11, 13, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37.

5.5 The Substitution Rule

- **Substitution Rule for Indefinite Integrals**

> Let g be a differentiable function and suppose that F is an antiderivative of f, then
> $$\int f\left(g(x)\right)g'(x)dx = F\left(g(x)\right) + C$$

- **Example** Evaluate $\displaystyle\int \sin 3x dx$.

- **Example** Evaluate $\displaystyle\int x \sin x^2 dx$.

- **Example** Evaluate $\displaystyle\int \sin^3 2x \cos 2x dx$.

- **Example** Evaluate $\displaystyle\int x^3\sqrt{x^4+11}dx$.

- **Example** Evaluate $\displaystyle\int \frac{x^5}{\sqrt{x^2+1}}dx$.

- **Example** Evaluate $\int \tan x dx$.

- **Example** Evaluate $\int \sec x dx$.

- **Substitution Rule for Definite Integrals**

Let g have continuous derivative on $[a,b]$, and let f be continuous on the range of g, then

$$\int_a^b f\left(g(x)\right) g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

where $u = g(x)$.

- **Example** Evaluate $\displaystyle\int_0^4 \sqrt{2x+1}dx$.

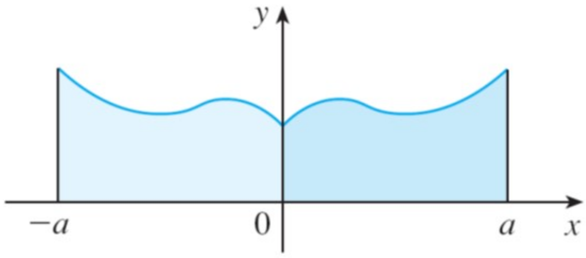
- **Example** Evaluate $\displaystyle\int_0^1 \frac{x+1}{(x^2+2x+6)^2}dx$.

- **Example** Evaluate $\int_{\frac{\pi^2}{9}}^{\frac{\pi^2}{4}} \frac{\cos\sqrt{x}}{\sqrt{x}}dx.$

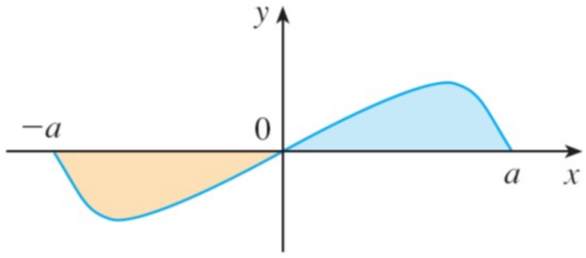
- **Example** Evaluate $\int_1^e \frac{\ln x}{x}dx.$

- **Symmetry Theorem**

If f is an even function, then
$$\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$$
If f is an odd function, then
$$\int_{-a}^{a} f(x)dx = 0$$



(a) f even, $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$



(b) f odd, $\int_{-a}^{a} f(x)\,dx = 0$

- **Example** Evaluate $\displaystyle\int_{-\pi}^{\pi} \cos\left(\frac{x}{4}\right) dx$.

- **Example** Evaluate $\displaystyle\int_{-5}^{5} \frac{x^5}{x^2+4} dx.$

- **Example** Evaluate $\displaystyle\int_{-2}^{2} \left(x \sin^4 x + x^3 - x^4\right) dx.$

- **Example** Evaluate $\displaystyle\int_{-\pi}^{\pi} \sin^3 x \cos^5 x dx.$

Exercises 5.5 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 37, 41, 43, 49, 51, 53, 55, 57, 59, 61, 63, 65, 67, 73.

6.1 Areas between Curves

- **Region Above the x-Axis**

> Let $y = f(x)$ be continuous and nonnegative on $a \leq x \leq b$, and let R be the region bounded by the curve $y = f(x)$, the lines $x = a$, $x = b$ and the x axis. The area of the region R is
>
> $$A = \int_a^b f(x)dx$$

- **Example** Find the area of the region R under $y = x^4 - 2x^3 + 2$ between $x = -1$ and $x = 2$.

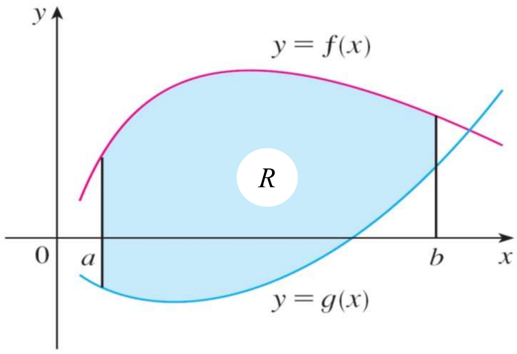
- **Region Below the x-Axis**

> Let $y = f(x)$ be continuous and non-positive on $a \le x \le b$, and let R be the region bounded by the curve $y = f(x)$, the lines $x = a$, $x = b$ and the x axis. The area of the region R is
>
> $$A = -\int_a^b f(x)dx$$

- **Example** Find the area of the region R bounded by $y = \dfrac{x^2}{3} - 4$, the x axis, $x = -2$ and $x = 3$.

- **A Region Between Two Curves**

> Consider the curves $y = f(x)$ and $y = g(x)$ with $g(x) \leq f(x)$ on $a \leq x \leq b$. Let R be the region bounded between the curves $y = f(x)$, $y = g(x)$ and the lines $x = a$ and $x = b$. The area of R is
>
> $$A = \int_a^b [f(x) - g(x)]\, dx$$

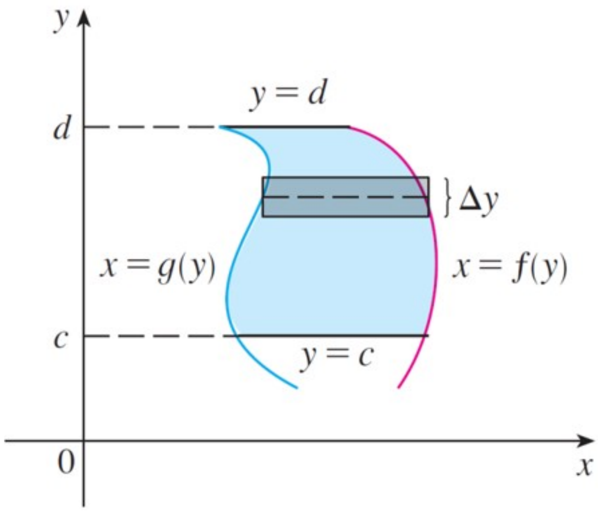


- **Example** Find the area of the region between the curves $y = x^2$ and $y = 2x - x^2$.

- **Horizontal Slicing**

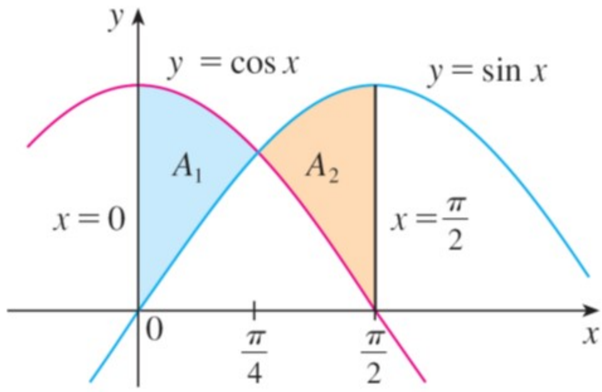
Consider the curves $x = f(y)$ and $x = g(y)$ with $g(y) \le f(y)$ on $c \le y \le d$. Let R be the region bounded between the curves $x = f(y)$, $x = g(y)$ and the lines $y = c$ and $y = d$. The area of R is

$$A = \int_c^d [f(y) - g(y)]\, dy$$



- **Example** Find the area of the region bounded between the parabola $y^2 = 4x$ and the line $4x - 3y = 4$.

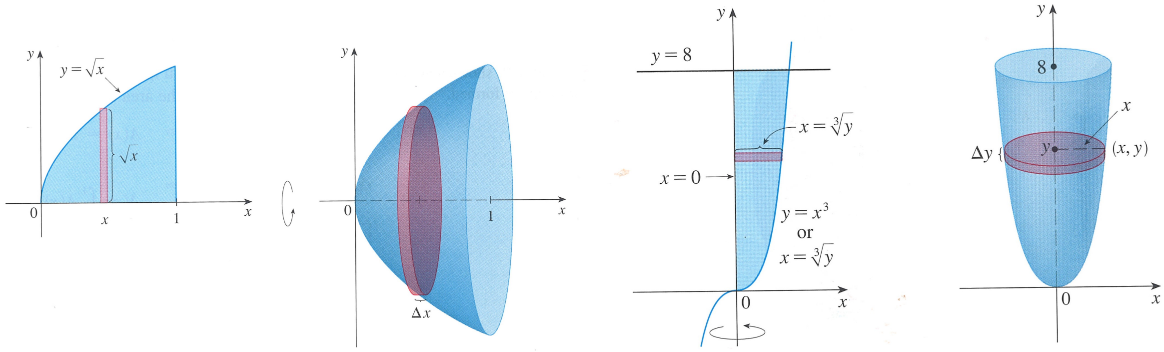
- **Example** Find the area of the region bounded by the curves $y = \sin x$, $y = \cos x$, $x = 0$ and $x = \dfrac{\pi}{2}$.



Exercises 6.1 1, 3, 7, 11, 13, 17, 21.

6.2 Volumes

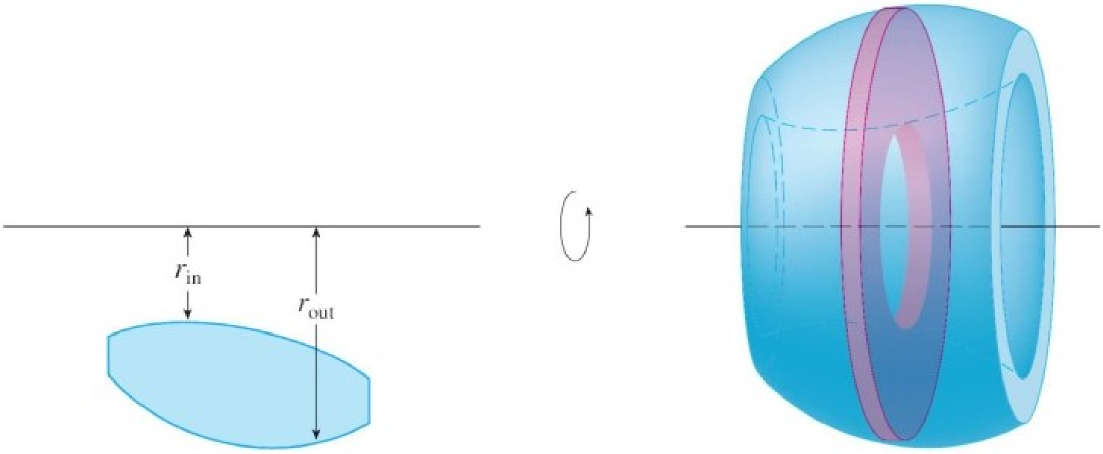
- When a plane region, lying entirely on one side of a fixed line in its plane, is revolved about that line, it generates a solid of revolution.



- **Example** Find the volume of the solid of revolution obtained by revolving the plane region R bounded by $y = \sqrt{x}$, the x-axis, and the line $x = 1$ about the x-axis.

- **Example** Find the volume of the solid generated by revolving the region bounded by the curve $y = x^3$, the y-axis and the line $y = 8$ about the y-axis.

- Sometimes, slicing a solid of revolution results in disks with holes in the middle. We call them washers.



- **Example** Find the volume of the solid generated by revolving the region bounded by the parabola $y = x^2$ and $y^2 = 8x$ about the x-axis.

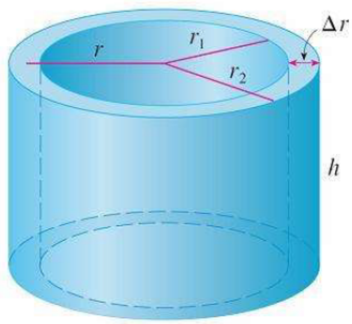
- **Example** The semicircular region bounded by the curve $x = \sqrt{4 - y^2}$ and the y-axis is revolved about the line $x = -1$. Set up the integral that represents its volume. Then compute this volume.

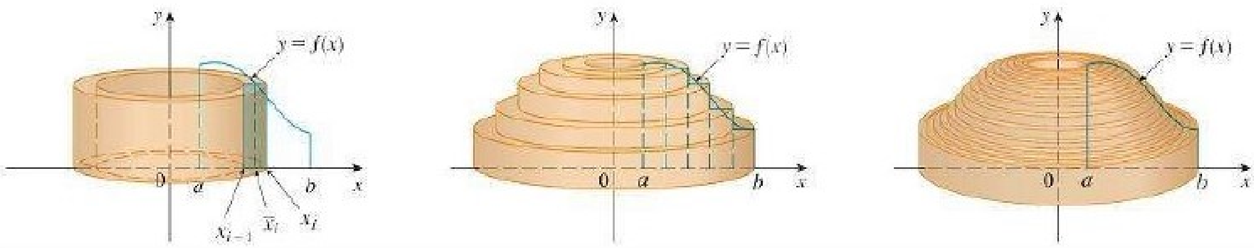
Exercises 6.2 1, 3, 5, 7, 9, 11, 13, 15, 17.

6.3 Volumes by Cylindrical Shells

- A cylindrical shell is a solid bounded by two concentric right circular cylinders. If the inner radius is r_1 and the outer radius is r_2, and the height is h, then the volume of the cylindrical shell is

$$V = \pi\left(r_2^2 - r_1^2\right)h$$





- **Example** The region bounded by $y = \dfrac{1}{\sqrt{x}}$, the x-axis, $x = 1$ and $x = 4$ is revolved about the y-axis. Find the volume of the resulting solid.

- **Example** Find the volume of the solid generated by revolving the region in the first quadrant that is above the parabola $y = x^2$ and below the parabola $y = 2 - x^2$ about the y-axis.

- **Example** Set up and evaluate an integral for the volume of the solid that results when the region R bounded by the curve $y = 3 + 2x - x^2$ and the two coordinate axes in the first quadrant is revolved about

  (a) the x-axis,

  (b) the y-axis,

  (c) the line $y = -1$,

  (d) the line $x = 4$.

Exercises 6.3 3, 5, 9, 11, 15, 17, 41.

6.5 Average Value of a Function

- **Average Value of Function**

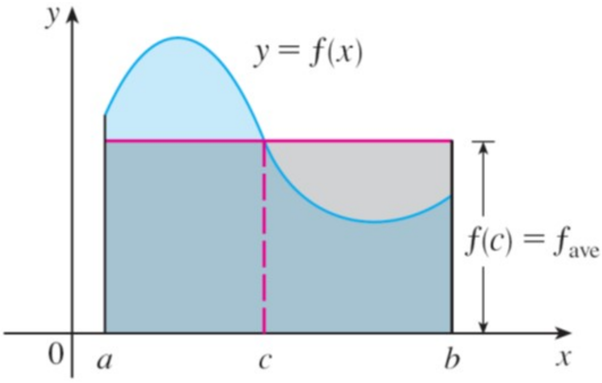
If f is integrable on the interval $[a, b]$, then the average value of f on $[a, b]$ is

$$\frac{1}{b-a}\int_a^b f(x)dx$$

- **Example** Find the average value of the function $f(x) = x \sin x^2$ on the interval $[0, \sqrt{\pi}]$.

- **Mean Value Theorem for Integrals**

> If f is continuous on $[a, b]$, then there is a number c between a and b such that
> $$f(c) = \frac{1}{b-a}\int_a^b f(x)dx$$



- **Example** Find all the values of c that satisfy the Mean Value Theorem for integrals for $f(x) = x^2$ on the interval $\left[-\sqrt{3}, \sqrt{3}\right]$.

Exercises 6.5 3, 5, 7, 9(a)(b), 13.

7.1 Integration by Parts

- **Integration by Parts**

$$\int u dv = uv - \int v du$$

$$\int_a^b u(x)v'(x)dx = [u(x)v(x)]_a^b - \int_a^b v(x)u'(x)dx$$

- **Example** Find $\int x \cos x dx$.

- **Example** Find $\int_1^2 \ln x dx$.

- **Example** Find $\int \sin^{-1} x dx$.

- **Example** Find $\int_1^2 t^6 \ln t dt$.

- **Example** Find $\displaystyle\int x^2 \sin x dx$.

- **Example** Find $\displaystyle\int e^x \sin x dx$.

Exercises 7.1 3, 5, 7, 9, 11, 13, 15, 19, 21, 23, 25, 31, 33 ($hint$: let $y = \sqrt{x}$).

7.2 Some Trigonometric Integrals

- **Example** Find $\displaystyle\int \sin^5 x dx$.

- **Example** Find $\displaystyle\int \cos^4 x dx$.

- **Example** Find $\displaystyle\int \frac{\sin^3 x}{\cos^4 x} dx$.

- **Example** Find $\displaystyle\int \sin^2 x \cos^4 x dx$.

- **Some Trigonometric Identities**

(a) $\sin mx \cos nx = \frac{1}{2}[\sin(m+n)x + \sin(m-n)x]$

(b) $\sin mx \sin nx = -\frac{1}{2}[\cos(m+n)x - \cos(m-n)x]$

(c) $\cos mx \cos nx = \frac{1}{2}[\cos(m+n)x + \cos(m-n)x]$

- **Example** Find $\int \sin 2x \cos 3x dx$.

- **Example** Find $\int \tan^5 x dx$.

- **Example** Find $\displaystyle\int \sec x \tan^3 x dx$.

- **Example** Find $\displaystyle\int \tan^6 x \sec^4 x dx$.

Exercises 7.2 1, 3, 5, 7, 11, 13, 21, 23, 25, 27, 29, 41, 43, 45 (*hint:* simplify the integrand).

7.3 Trigonometric Substitutions

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Expression	Substitution	Domain	Identity
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{a^2 + x^2}$	$x = a\tan\theta$	$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	$1 + \tan^2\theta = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$0 \leq \theta \leq \pi,\ \theta \neq \frac{\pi}{2}$	$\sec^2\theta - 1 = \tan^2\theta$

- **Example** Find $\int \sqrt{a^2 - x^2}dx$.

- **Example** Find $\displaystyle\int \frac{dx}{\sqrt{9+x^2}}$.

- **Example** Find $\displaystyle\int_2^4 \frac{\sqrt{x^2-4}}{x}dx$.

- **Example** Find $\displaystyle\int \frac{dx}{\sqrt{x^2+2x+26}}$.

- **Example** Find $\displaystyle\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2+9)^{3/2}}dx$.

Exercises 7.3 5, 7, 9, 11, 13, 17, 21, 23, 25, 27, 29.

7.4 Integration of Rational Functions Using Partial Fractions

- **Example** Find $\displaystyle\int \frac{2}{(x+1)^3}dx.$

- **Example** Find $\displaystyle\int \frac{x^3+x}{x-1}dx.$

- **Example** Find $\displaystyle\int \frac{2x+2}{x^2-4x+8}dx.$

- **Example** Decompose $\dfrac{3x-1}{x^2-x-6}$ into partial fractions and find $\displaystyle\int \frac{3x-1}{x^2-x-6}dx$.

- **Example** Find $\int \frac{x}{(x-3)^2}dx$.

- **Example** Find $\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx.$

- **Example** Find $\displaystyle\int \frac{6x^2 - 3x + 1}{(4x+1)(x^2+1)}dx.$

- **Example** Write out the form of the partial fraction decomposition of the function

$$\frac{x^3+x^2+1}{x(x-1)(x^2+x+1)(x^2+1)^3}$$

- **Example** Evaluate $\displaystyle\int \frac{\sqrt{x+4}}{x}dx$.

Exercises 7.4 1, 3, 5, 7, 9, 11, 15, 17, 21, 23, 25, 27, 29, 31.

7.8 Improper Integrals

Infinite Intervals

- Integrals of the form $\int_0^\infty \frac{1}{1+x^2}dx$, $\int_{-\infty}^{-1} xe^{-x^2}dx$ and $\int_{-\infty}^{\infty} x^2e^{-x^2}dx$ are called improper integrals with infinite limits.

- Consider $\int_0^\infty e^{-x}dx$.

- **Definition**

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{a\to-\infty}\int_a^b f(x)\,dx,$$
$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty}\int_a^b f(x)\,dx$$

If the limits on the right exists and have finite values, we say that the corresponding improper integrals converge and have those values. Otherwise, we say that the improper integrals diverge.

- **Example** Find, if possible, $\displaystyle\int_{-\infty}^{-1} xe^{-x^2}dx$.

- **Example** Find, if possible, $\displaystyle\int_{0}^{\infty} \sin x dx$.

Both Limits Infinite

- **Definition**

If both $\int_{-\infty}^{a} f(x)dx$ and $\int_{a}^{\infty} f(x)dx$ converge, then $\int_{-\infty}^{\infty} f(x)dx$ is said to converge and have value

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{a} f(x)dx + \int_{a}^{\infty} f(x)dx$$

Otherwise, $\int_{-\infty}^{\infty} f(x)dx$ diverges.

- **Example** Determine whether $\int_{-\infty}^{\infty} \frac{1}{1+x^2}dx$ is convergent. If yes, find its value.

- **Example** Determine whether $\int_{-\infty}^{\infty} \frac{x}{1+x^2}dx$ is convergent. If yes, find its value.

- **Example** For what values of p does the integral $\displaystyle\int_1^\infty \frac{1}{x^p}dx$ converge?

Discontinuous Integrands

- **Example** Evaluate $\displaystyle\int_{-2}^{1} \frac{1}{x^2} dx$.

Integrands Infinite at an End Point

- **Definition**

Let f be continuous on the half-open interval $[a, b)$ and suppose that $\lim\limits_{x \to b^-} |f(x)| = \infty$, then

$$\int_a^b f(x)\, dx = \lim_{t \to b^-} \int_a^t f(x)\, dx$$

provided that this limit exists and is finite, in which case we say that the integral converges. Otherwise, we say that the integral diverges.

Let f be continuous on the half-open interval $(a, b]$ and suppose that $\lim\limits_{x \to a^+} |f(x)| = \infty$, then

$$\int_a^b f(x)\, dx = \lim_{t \to a^+} \int_t^b f(x)\, dx$$

provided that this limit exists and is finite, in which case we say that the integral converges. Otherwise, we say that the integral diverges.

- **Example** Evaluate, if possible, the improper integral $\displaystyle\int_0^2 \frac{dx}{\sqrt{4-x^2}}$.

- **Example** Evaluate, if possible, $\displaystyle\int_0^{16} \frac{1}{\sqrt[4]{x}}dx$.

- **Example** Evaluate, if possible, $\displaystyle\int_0^1 \frac{1}{x}dx$.

- **Example** For what values of p does the integral $\displaystyle\int_0^1 \frac{1}{x^p} dx$ converge?

- **Example** Evaluate, if possible, $\displaystyle\int_0^1 \ln x dx$.

Integrands Infinite at an Interior Point

- **Definition**

Let f be continuous on $[a, b]$ except at a number $c \in (a, b)$, and suppose that $\lim_{x \to c} |f(x)| = \infty$. Then we define

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)dx$$

provided both integrals on the right converge. Otherwise, we say that $\int_a^b f(x)dx$ diverges.

- **Example** Show that $\int_{-2}^{1} \frac{1}{x^2}dx$ diverges.

- **Example** Evaluate, if possible, the improper integral $\displaystyle\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}}$.

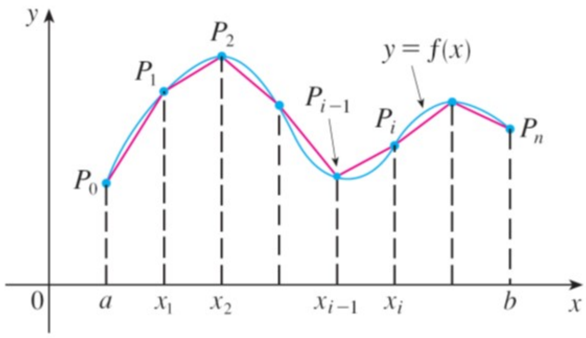
Exercises 7.8 5, 9, 11, 13, 15, 17, 21, 23 (*hint*: let $x^3 = 3\tan\theta$), 25, 27, 29, 31, 37 (*hint*: let $u = e^x$).

8.1 Arc Length

- **Arc Length**

If f' is continuous on $[a, b]$, the **arc length** of the curve $y = f(x)$, $a \leq x \leq b$ is given by

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$



- **Example** Find the length of the arc of the curve $y = x^{\frac{3}{2}}$ from the point $(1, 1)$ to the point $(4, 8)$.

-

If g' is continuous on $[c, d]$, the **arc length** of the curve $x = g(y)$, $c \leq y \leq d$ is given by

$$\int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

- **Example** Find the length of the arc of the parabola $y^2 = x$ from $(0, 0)$ to $(1, 1)$.

Exercises 8.1 5, 7, 9, 11, 37.

8.2 Area of Surface of Revolution

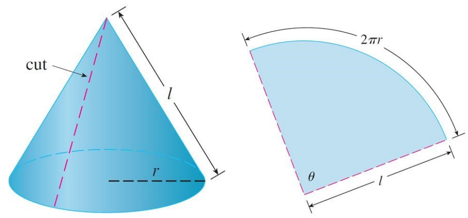
- **Surface Area**

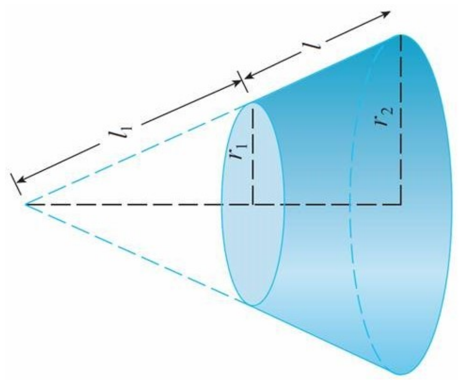
The **area of the surface** of revolution generated by rotating the curve $y = f(x)$, $a \leq x \leq b$ about the **x-axis** is

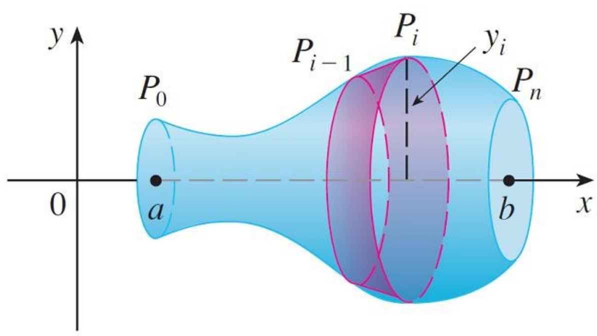
$$S = 2\pi \int_a^b y(x)ds = 2\pi \int_a^b y(x)\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

The **area of the surface** of revolution generated by rotating the curve $y = f(x)$, $a \leq x \leq b$ about the **y-axis** is

$$S = 2\pi \int_a^b x ds = 2\pi \int_a^b x\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$







- **Example** Find the area of the surface of revolution generated by revolving the curve $y = \sqrt{4 - x^2}$, $-1 \le x \le 1$ about the x-axis.

- **Example** The arc of the curve $y = x^2$ from $(1, 1)$ to $(2, 4)$ is rotated about the y-axis. Find the area of the resulting surface.

Exercises 8.2 5, 7, 13.